

ASSIGNMENT - BUILDING A WINDOWS LIVE DATA ACQUISITION TOOLKIT

OVERVIEW

At times, a forensic investigator will need to build their own tools to solve a simple or complex problem. For this assignment, you will need to add the proper PowerShell commands to the template script file for performing a live data acquisition on a Windows 10 or 11 system. The PowerShell commands must meet all requirements including filtered fields and output requirements (e.g. text, CSV, JSON). Secondly, you must write proper documentation for your tool. Using the provided template, complete all sections and include screenshots and all information needed for your new tool. **It is highly recommended** that you take the viewpoint of a person that has never ran a PowerShell tool while developing your documentation.

INSTRUCTIONS



There is no case number for this assignment

Review both the **ldat-template.ps1** and **tool-documentation-template.docx**. You are required to name your tool (e.g. Windows10-LiveDataAcquisitionToolkit). It is highly recommended you use VSCode to update and run the script.

Script Development

- Run the template BEFORE YOU BEGIN YOUR EDITS!
- Update the top of the PowerShell script. This is important data including purpose, usage, author, etc.
- Update the name of the tool
- Complete the following steps under the “#———LOOK HERE (STEP X)———” sections. **ONLY update these sections!**
 - Step 1: Check to make sure script is run as administrator
 - Step 2: Obtain computer system information using a PowerShell cmdlet. Output must be in text format
 - Step 3: Obtain a list of running processes including their name and ID. Output must be in CSV format
 - Step 4: Obtain a list of services with name, display name, status, start type, and service type. Output must be in text format
 - Step 5: Obtain a list of local users with their group memberships. Output must be in text format
 - Step 6: Obtain a list of local and remote shares. Output must be in text format
 - Step 7: Obtain a list of network adapters. Output must be in JSON format
 - Step 8: Obtain the IP network configurations for all adapters. Output must be in text format
 - Step 9: Obtain a list of TCP connections including local address, local port, remote address, remote port, state, and owning process. Output must be in text format
 - Step 10: Obtain a list of UDP connections including local address, local port, remote address, remote port, state, and owning process. Output must be in text format
 - Step 11: Obtain the ARP table. Output must be in text format
 - Step 12: Obtain the Route table. Output must be in text format
 - Step 13: Hash all output files using SHA256, include the path and hash, and put them into a file named `TIMESTAMP_CASENUMBER_HOSTNAME-Filehashes.txt`

Documentation

- Complete all required sections
- Update the name of the tool

REQUIRED SOFTWARE

Name	Vendor
PowerShell	Microsoft

Name	Vendor
VSCode	Microsoft

REQUIRED FILES

- ldat-template.ps1
- tool-template.docx

DELIVERABLE REQUIREMENTS

- Working version of your script
- Tool Documentation
 - Purpose
 - Legal Notice
 - Requirements
 - Tool Usage
 - Output Details
 - Side Effects
 - Contact Information

GRADING RUBRIC

- PowerShell Tool - 50pts
 - Works properly with no errors - 20 pts
 - Header documentation complete - 6pts
 - All steps completed - 24pts total
 - Proper command & filters - 1pts
 - Proper output - 1pts
- Documentation - 50pts
 - Used template - 10pts
 - All sections filled out adequately - 40pts
 - Includes screenshots
 - Includes proper usage
 - Includes all side effects